

Aliah University

Odd (Autumn)-Semester Examination – 2025

(For 7th Semester BTech(CSE) Programme)

Paper Name: Internet of Things

Paper Code: CSEUGPE12

Full Marks: 80

Time: 3 Hrs

Group – A

(5X2=10)

1. Give a one-sentence answer – Remember, CO1-CO5
- What is a node in a sensor network?
 - Explain the significance of interoperability in the context of IoT.
 - State the motivation behind Fog Computing.
 - What is a Raspberry Pi?
 - What is the role of an actuator in IoT?

Group – B

(Answer all questions)

(6X5=30)

2. Compare IoT and M2M communication systems. **Understand, CO1**
3. What is Machine-to-Machine (M2M) communication? Discuss its role in IoT. 2+3
Understand, CO1
4. Explain how sensors and actuators are interfaced with Arduino. **Understand, CO2**
5. Describe the basic steps to integrate a temperature sensor with Arduino. Include the circuit setup and code snippet. **Apply, CO2**
6. Explain the concept of interoperability in IoT with example. Why is it important? 2+3
Understand, CO3
7. Explain the concept of Connected Vehicles. **Understand, CO3**

Group – C

(Answer any 4 questions)

(10X4=40)

8. a) Explain the architecture of an IoT-based smart home system.
- b) Discuss how fog computing enhances IoT applications. 5+5 **Understand, CO4**
9. Write an Arduino code for a smart home application which will constitute of a Heater, Fan and Air Conditioner. Describe the code in details. 5+5 **Apply, CO2**

10. Discuss the applications of IoT in the following fields:
a) Healthcare
b) Agriculture **5+5 Understand, CO5**
11. Write a detailed note on the role of IoT in Smart Cities. Discuss any two use cases.
5+5 Understand, CO4
12. a) Compare Cloud Computing and Fog Computing in context of IoT.
b) Describe the architecture of a Smart Home system. **5+5 Understand, CO4**
13. Explain the role of IoT in Healthcare monitoring systems. **5+5 Understand, CO5**
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